Cardiac Deformation Lab

Image-Based Cardiac Deformation Mapping & Kinematic Strain Analysis

Analysis ID:	831fa382-3a1c-431c-aa00-23c7c84b6c26	**Analysis Type:**	Multi-Frame Temporal Sequence Analysis
Execution Date:	October 16, 2026 at 04:43:48 UTC	**Motion Model:**	Dense Farnebäck Optical Flow
Spatial Calibration:	Not provided; displacement values reported in image pixels (px)	**Calibration Status:**	Image-Space Only
Analysis Region:	Full Image (All image-space pixels evaluated)	**Region Scope:**	Full Image
Frames Analyzed:	4	**Consecutive Pairs:**	3

1. Displacement & Kinematic Motion

Metric Parameter	Value (Pixel Units)	Description
Mean Displacement	1.2984 px	Average Euclidean magnitude of displacement vector field
Maximum Displacement	11.4322 px	Peak regional apparent motion amplitude across the field
Minimum Displacement	0.0000 px	Lowest regional motion magnitude observed
Displacement Std. Dev.	0.2300 px	Spatial heterogeneity of tissue displacement

2. Small-Deformation Strain Tensor & Principal Fields

Strain Component	Mean	Min	Max	Std Dev
Normal Exx (du/dx)	N/A	N/A	N/A	N/A
Normal Eyy (dv/dy)	N/A	N/A	N/A	N/A
Shear Exy (0.5*(du/dy+dv/dx))	N/A	N/A	N/A	N/A
Principal Strain 1 (Major)	N/A	N/A	N/A	N/A
Principal Strain 2 (Minor)	N/A	N/A	N/A	N/A

3. Advanced Deformation Metrics & Geometric Projection

Metric Parameter	Observed Value	Physical / Mathematical Interpretation
Motion Coherence	N/A	Cosine consistency of motion vectors within local 5x5 spatial kernels
Strain Gradient Magnitude	N/A	Spatial rate of change of principal strain fields
Strain Anisotropy Index	N/A	Directional preference of tissue deformation (|e1 - e2| / (|e1| + |e2|))

Strain Entropy	N/A	Shannon entropy describing dispersion of regional strain histograms
Deformation Energy	N/A	Pseudo-elastic strain energy density surrogate (0.5 * Tr(eps * eps))
Radial Deformation (Mean)	N/A	Motion projection along radial axes from ventricular centroid approximation
Circumferential Deformation (Mean)	N/A	Motion projection orthogonal to radial vectors (tangential shortening)

4. Multi-Frame Temporal Strain Trajectory

Temporal Parameter	Value	Mathematical Definition
Temporal Mean Principal Strain	0.0616	Average principal strain magnitude across all consecutive frame pairs
Temporal Std Dev	0.0090	Standard deviation of frame-to-frame strain trajectory
Trajectory Range (Min - Max)	0.0490 to 0.0694	Minimum and peak principal strain observed during cine sequence
Temporal Stability Index (TSI)	0.8728	Smoothness metric 1 / (1 + variance) measuring trajectory stability

5. Cardiac Deformation Fingerprint (10-Dimensional)

Dimension Symbol	Canonical Name	Value
D_mean	Mean Displacement	N/A
D_max	Maximum Displacement	N/A
epsilon_mean	Representative Mean Strain	N/A
epsilon_max	Representative Max Strain	N/A
epsilon_principal	Representative Principal Strain	N/A
AI	Strain Anisotropy Index	N/A
H	Strain Entropy	N/A
G_epsilon	Strain Gradient Magnitude	N/A
C	Motion Coherence	N/A
TSI	Temporal Stability Index	N/A

7. Technical Interpretation & Research Prototype Disclaimer

Methodological Scope: The metrics, vector fields, and spatial colormaps detailed in this report represent apparent image-space deformation computed via Farnebäck dense optical flow and classical continuum strain tensor decomposition. Measurements are expressed in image coordinate units (pixels) and relative mathematical differentials.

RESEARCH ONLY — NO CLINICAL DIAGNOSTIC ASSERTIONS:
This software is an experimental computational research prototype. All values, including the Cardiac Deformation Fingerprint (10-D) and Temporal Stability Index (TSI), are algorithmic descriptors of pixel kinematics. This system does NOT provide medical diagnoses, clinical risk assessments, normal/abnormal classifications, or treatment recommendations. This report must NOT be used as a medical device or diagnostic decision-making instrument.